

SURVEY SAMPLING AND SAMPLING

DISTRIBUTION

Sampling:-

Sampling is a statistical technique which is used in almost every field in order to collect information & on the basis of this information inferences about the characteristics of a population are made.

Parameters:-

The values of the population characteristics are summarized by certain numerical descriptive measures.

Statistic:

The observations composing a sample are used to calculate a corresponding numerical descriptive measures.

Statistical Populations:-

It is defined as the aggregate or totality of all individual members or objects, whether-animate, or inanimate, concrete or abstract of some characteristics of interest.

Sampling units:

The individual members of the population are called sampling units or simply units. A sampling units from which information is required may be a college student, an animal etc.

Sample of size n :-

A set of n sampling units selected from a given population.

Population distribution:-

The numerical values assigned to units of interest are treated as values of a random variable X and the distribution of X is called the population distribution.

Size of population:

The total number of units in a finite population is the size of the population & is denoted by " N ". The example of infinite population are the population of all points on a line, the population of pressures of various points in the atmosphere, etc.

Existent population & hypothetical population:-

A population of concrete units such as trees, households, students is called an existent population. While a hypothetical population consists of all conceivable ways in which an event can occur e.g. all possible throws of a dice. Such a population does not exist in a concrete manner but is only to be thought of.

Sample population:-

A sampled population is that from which a sample is chosen.

Example: The sample population may consist of patients in district hospitals.

Target population:-

A population about which we wish to draw inferences is called a target population.

Example: It may consist of the total number of patients in the province.

Discrete & continuous population:-

A population is discrete when the number of units comprising the population is countable otherwise it is continuous.

Basic purpose of sampling:-

- (i) to provide sufficient information about the characteristic of the population
- (ii) to find reliability of the estimates derived from the sample.

Advantages of Sampling:-

- (1) Sampling saves money as it is much cheaper to collect the desired information from a small sample than from the whole population.
- (2) Sampling saves a lot of time & energy.
- (3) It provides information that is almost accurate.
- (4) Sampling makes it possible to obtain more detailed information from each unit of the sample.
- (5) Sampling is essential to obtaining the data when the measurement process physically damage or destroys the sampling unit under investigation.
- (6) Sampling may be the only means available for obtaining the needed information when the population appears to be infinite such as the population of mountains.
- (7) Sampling is extensively used to obtain some of the census information.
- (8) The most important advantage of sampling is that it provides a valid measure of reliability for the sample estimates & this is ^{one} of the two basic purposes of sampling.

Sample design:-

A sample design is a definite statistical plan concerned with all principal steps taken in the selection of a sample and the estimation procedure.

Sample survey:-

The term survey has been defined as a means of collecting information to meet a definite need. When a survey is carried out by a sampling method it is called a sample survey.

Main steps in a sample survey:-

- i) clearly state the objectives of the survey.
- ii) define the population we wish to study as clearly as possible.
- iii) construct the sampling frame by clearly defining the sampling units.
- iv) choose an appropriate sample design & proper sample size.
- v) organize a reliable field work to achieve the objectives of the survey.
- vi) summarize and analysis the data.

Sampling frame:

A sampling frame is a complete list or a map that contains all the N sampling units in a population.

The requirements of a reasonably good frame are that the frame should

- i) not contain inaccurate sampling units
- ii) be complete & exhaustive
- iii) be free from errors of duplication
- iv) be as up-to-date

Probability Sampling:-

When each unit in a population has a non zero probability of its being included in the sample the sampling is said to be probability sampling. It is also called a random sampling.

Types of probability sampling:-

- ① Simple Random Sampling
- ② Stratified Random
- ③ Systematic sampling
- ④ cluster sampling

Advantage of Probability Sample:-

It provides a value estimate of sampling error.

It is used in various areas. agriculture, business

Non Probability Sampling:-

A non probability sampling also called non random sampling is a process in which the personal judgement determines which units of population are selected for sample.

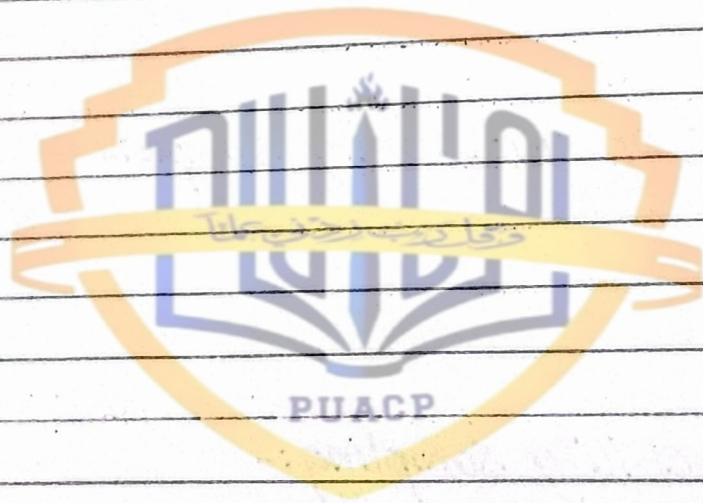
Disadvantage of probability sampling:

The reliability of ~~prob~~ sample results cannot be determine in terms of probability.

Non probability sampling techniques includes ~~prob~~ purpose sampling & quota sampling.

Sampling Bias:-

In survey sampling, the word bias means a systematic component of error which deprives a survey results of its representativeness. Bias is different from a random error in the sense that the random errors balance out in the long run while bias is accumulative & does not become less as the sample size increases.



"Systematic Sampling"

Systematic sampling is operationally more convenient than simple random sampling. This selection procedure is different from simple random sampling procedure in the context that simple random sampling every unit is selected randomly where as in the systematic sampling only first unit is selected at random and remaining units are automatically determinate by skip interval.

Suppose that there are N units in population number from one to N if $k = \frac{N}{n}$ the population of N units

would be divided into small n groups each of containing k units. Systematic Sampling is a technique to select a sample according to a system.

Systematic Sampling required:

- (1) numbering of all sampling units in the population that is i.e. $1, 2, \dots, N$
- (2) Decide the value of k where k is the length of the interval as $k = \frac{N}{n}$
- (3) Subdivided the population into n groups each of the size k .
- (4) Draw a random sample between 1 and k and every k th units these after with be included in the sample.

Example:

$$N=20 \quad n=4$$

$$k = \frac{N}{n} = \frac{20}{4} = 5$$

1, 2, 3, 4, 5

6, 7, 8, 9, 10

11, 12, 13, 14, 15

16, 17, 18, 19, 20

$\begin{array}{cccc} 1-k & k+1-2k & 2k+1-3k & 3k+1-4k \\ \hline 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 \end{array}$

all sampling units in interval 1 has equal probability. select a random number suppose that the number selected is u from the remaining sampling units are selected as $u, u+k, u+2k, u+3k$. All are in relative position selection of whole dependent on the selection of first units. It has random start

(the random number chosen from the first group of k units).

Compare with stratification:-

Stratification has so many positive points each stratum of size k small $n_1=1$ or $n_2=1$ within groups selection change. In stratified sampling relative position change which represent more precise than systematic sampling.

$$2, 7, 12, 17$$
$$[2, 2+k, 2+2k, 2+3k]$$

In systematic sampling, if it is too low in group one then it will be too low in remaining groups and if it is too high in group one then it will be too high in all groups.

for example class roll number:-
population lying in some order.

Advantages:-

It is convenient, easy, cheap.
and less expensive or time consuming.

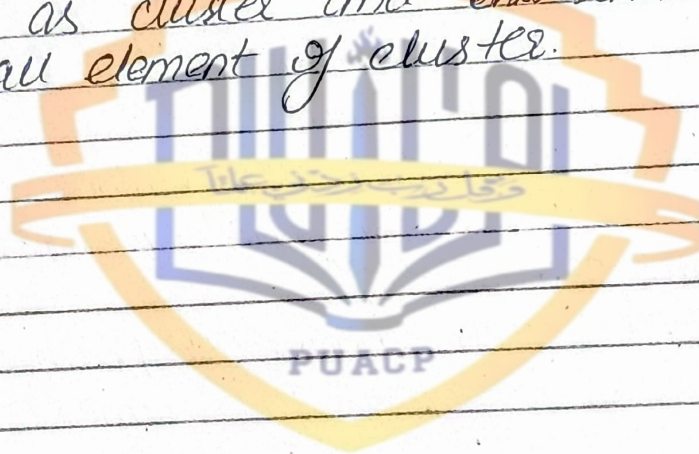
In some situations it gives biased sample $k = \frac{N}{n}$

$$N = kn$$

k is a multiple integral of n at k . If k is odd, then if first sampling unit lies in centre of first group then all will be in centre and this is called centrally located sample. If k is an even then $\frac{k}{2}$ -th and $\frac{k}{2}+1$ -th both are centrally located sample.

* Relation of systematic sampling to cluster sampling:

Cluster is a larger group of elements which is further divisible into smaller units of elements. For example income of a family is a cluster & income of a person is a small element. In systematic sampling population is divided into k systematic sample each contain " n " units. We can consider the k systematic sample as cluster and n units as a small element of cluster.



(Define the term accuracy & precision (12)
with reference to sampling theory:

You can never make exact measurement in an experiment. How far away from the "mark" you are is described by accuracy & how well you measure is described by precision.

Accuracy:-

Accuracy is how close you are to the true value for example: Let's say you know your true height is exactly is 5'9"

- you measure yourself with a yard stick and get 5'0" your measurement is not accurate
- you measure again yourself with a laser yard stick and get 5'9" your measurement is accurate.

* The true value is sometimes called the theoretical value.

Precision:-

Precision is how close two or more measurements are to each other. If you consistently measure your height as 5'0" with a yard stick, your measurements are precise.

* Accuracy of analysis & precision together

If you are precise, that doesn't necessarily mean you are accurate. However, if you are consistently accurate, you are also precise.

* "More precise"

If you want to tell which set of data is more precise, measure the range.

for example:

Two sets of data:

• Sample A: 32.56, 32.55, 32.48, 32.48, 32.49

• Sample B: 15.38, 15.37, 15.36, 15.37, 15.36

Subtract lowest data point from highest:

$$\bullet \text{ Sample A: } 32.56 - 32.48 = 0.08$$

$$\bullet \text{ Sample B: } 15.38 - 15.32 = 0.06$$

Sample B has the lowest range (0.06) and so is the more precise.

Ratio Estimator (method of estimation)

frequently, the quantity that is to be estimated from a simple random sample is the ratio of two variables, both of which varies from unit to unit. The one variable is small y , the variable of interest and the other variable is x , the scaling variable or auxiliary variable. Both of variables measures on the same sampling units

family	income y_i	Expenditure (x_i)
1	y_1	x_1
2	y_2	x_2
⋮	⋮	⋮
n	y_n	x_n

village	wheat crop	
	2008 currently y_i	1998 previous (x_i)
1	y_1	x_1
2	y_2	x_2
⋮	⋮	⋮
n	y_n	x_n

The

The population parameter to be estimated is the ratio

$$R = \frac{\text{total income}}{\text{total expenditure}} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i} = \frac{\bar{Y}}{\bar{X}}$$

Corresponding sample estimate is

$$\hat{R} = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i} = \frac{\bar{y}}{\bar{x}}$$

Ratio estimator is used:

When the sampling unit (household) comprises a group or cluster of elements (members) and our interest is in the population mean per element that is i.e. ratio estimator provides us the average per element in this situation.

* Sampling distribution of $\hat{R}$:

Sampling distribution of $\hat{R}$ is more complicated than that of $\bar{Y}$ because both the numerator $\bar{Y}$ & the denominator $\bar{X}$ vary from sample to sample. In sample samples in distribution of $\hat{R}$ is skew and $\hat{R}$ is usually a slightly biased estimate of R .

$$E(\hat{R}) \neq R$$

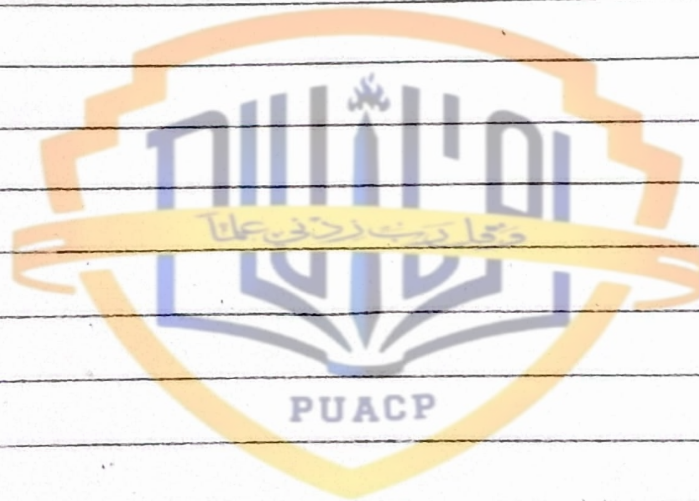
for $R = \frac{\bar{Y}}{\bar{X}}$ the estimator is $\hat{R} = \frac{\bar{y}}{\bar{x}}$

an estimator for population mean $\bar{Y}$ is

$$\bar{Y}_R = R \bar{X}$$

an estimator for population total Y is

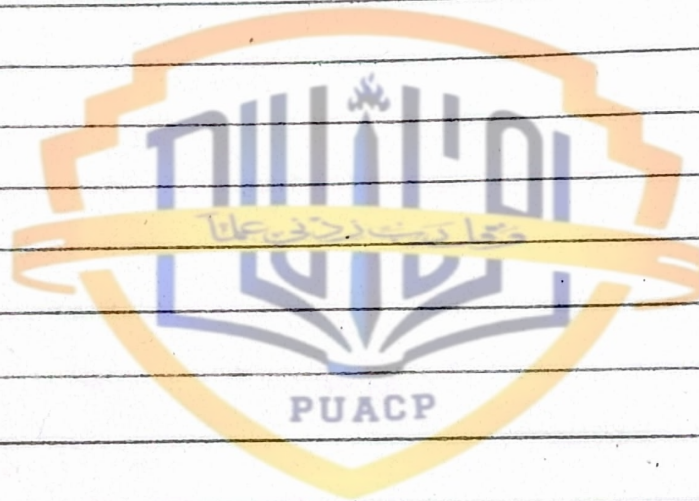
$$\hat{Y}_R = R X$$



Exact amount of bias: or Bias
using Hartley Ross approach:

Hartley Ross suggested a technique
which gives an exact amount of bias.
It also gives an upper bound to the
ratio of the bias to the standard error.
It is derived from covariance of the
qualities

$\hat{R}$ and $\bar{x}$



Ratio estimate in stratified random variable

for estimating population total Y , there are two ways of ratio estimate, define for stratified random sampling.

- ① Separate ratio estimate $\hat{Y}_{RS}$
- ② Combined ratio estimate $\hat{Y}_{RC}$

If R_h varies greatly from stratum to stratum then separate ratio estimate is used.

If the values of R_h close to each other the combined ratio estimate is use.

$$\bar{y}_{st} = \sum W_h \bar{y}_h$$

$$\hat{y}_{st} = \sum N_h \bar{y}_{st}$$

$$R_h = \frac{\bar{y}_h}{\bar{x}_h}$$

$$\hat{R}_h = \frac{\bar{y}_h}{\bar{x}_h} = \frac{y_h}{x_h}$$

$$\hat{Y}_{RS} = \sum \hat{Y}_{Rh}$$

$$\hat{Y}_R = \hat{R} X$$

$$\hat{Y}_{RC} = \frac{\bar{y}_{st}}{\bar{x}_{st}} X$$

$$= \frac{\hat{y}_{st}}{\bar{x}_{st}} X$$

$$\hat{Y}_R = \hat{R} X$$

$$\hat{Y}_R = \frac{\bar{y}}{\bar{x}} X$$

$$\hat{Y}_{RC} = \frac{\bar{y}_{st} X}{\bar{x}_{st}}$$